KLH UNIVERSITY

Koneru Lakshmaiah Education Foundation
(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)
Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in

STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	**I-III** Semester(2025-26)
Faculty Name	Ms. Md Razia
Student Name	Adunuri Bhavana Sri
Roll Number	2520030232
Branch	CSE
Course Outcome	1

Case Study Topic: HealthNet – Intelligent Hospital Resource Management System using BST

Work Done Summary:

- Implemented a Binary Search Tree (BST) in Java keyed by unique patient/ward IDs to index hierarchical medical information structure.
- Inserted multiple acute care admission entries and resource registries into the tree to execute seamless sorting patterns.
- Implemented emergency triage categorization, bed/ventilator reservations, medical lookup, profile record removals, and system integrity logs.
- Deleted discharged, incorrect, or cancelled emergency records while dynamically updating balance pointers to maintain correct tree properties.
- Utilized in-order tree traversals to generate list summaries of current cases structured by active priority levels sequentially.
- Compared hierarchical tree structures with traditional sequential lookup blocks (like ArrayLists), revealing massive timeline cuts for emergency records handling.

Implementation Details:

- **1. Created Node infrastructure class** with fields: patientId, severityScore, requiredResource, left, and right pointers.
- **2. insert()** applies recursive tree operations to add incoming triage files into exact hierarchical slots.

- **3. search()** evaluates specific patient identification strings quickly to verify ongoing operational status metrics.
- **4. delete()** eliminates records from active sorting nodes when patients are successfully moved out of emergency lanes.
- **5. inorderTraversal()** parses the structural path to present a perfectly sorted array of ongoing clinical cases.
- **6. Main simulation method** demonstrates adding records, checking specific profiles, execution of node dismissals, and final diagnostic log reports.

Result: Screenshots/Output

```
PS D:\JavaPrograms> javac HealthNetBST.java
PS D:\JavaPrograms> java HealthNetBST

HealthNet - Emergency Hospital Resource Index (BST Inorder Traversal):
Patient ID: 20 -> Severity: Low Alert -> Resource: General Ward
Patient ID: 30 -> Severity: Moderate Alert -> Resource: Oxygen Bed
Patient ID: 40 -> Severity: High Critical -> Resource: ICU Ventilator
Patient ID: 50 -> Severity: High Critical -> Resource: ICU Ventilator
Patient ID: 60 -> Severity: Acute Trauma -> Resource: Operation Theater

Searching Index for Patient ID 40...
Patient ID 40 Found successfully in the active care index.

Deleting Discharged Patient ID 20...

Updated BST Inorder Traversal After Removal:
Patient ID: 30 -> Severity: Moderate Alert -> Resource: Oxygen Bed
Patient ID: 40 -> Severity: High Critical -> Resource: ICU Ventilator
Patient ID: 50 -> Severity: High Critical -> Resource: ICU Ventilator
Patient ID: 60 -> Severity: Acute Trauma -> Resource: Operation Theater

System Operational Audit Complete: No data faults.
```

Time Complexity & Empirical Analysis:

- BST Record Insertion / Search / Removal: O(log n) average execution runtime.
- Full Tree Sorted Enumeration: O(n) sequential path traversal.
- Compared to linear arrays where searching takes O(n) runtime complexity, the hierarchical BST structure provides highly rapid indexing for fast clinical triage tracking.

GitHub Repository Link:
https://github.com/bhavana-sri25/Sketch-DSA-CO1

Student Signature **Faculty Signature**